

Project Planning Document

Agile Sprint Planning for Data Analysis Project

1. Introduction

Project planning is an essential phase in software development that helps teams organize tasks, allocate resources, and ensure timely delivery of project goals. This project follows the **Agile methodology**, where development work is divided into short iterations known as **Sprints**. Each sprint focuses on completing a set of predefined tasks to gradually build the final system.

The tasks are organized into **Epics**, **User Stories**, and **Story Points** to estimate effort and track progress efficiently.

2. Key Agile Concepts

Sprint

A **Sprint** is a fixed time period during which a development team works to complete a defined set of tasks. At the end of each sprint, a working part of the project should be completed.

Epic

An **Epic** represents a large task or major feature of the project that cannot be completed within a single sprint. Epics are divided into smaller units called **User Stories**.

User Story

A **User Story** is a small unit of work that represents a specific functionality or task. Multiple user stories together form an epic.

Story Point

A **Story Point** represents the **effort required to complete a user story**. It is not measured in time but rather in complexity, risk, and workload.

Story points often follow the **Fibonacci sequence**:

Story Point	Task Difficulty
1	Very Easy
2	Easy / Normal
3	Moderate
5	Difficult

3. Sprint Planning

Sprint 1

Epic 1: Data Collection

User Story ID	Task	Story Points
USN1	Gathering Data	2
USN2	Loading Data	1

Epic 2: Data Preparation

User Story ID	Task	Story Points
USN3	Handling Missing Values	3
USN4	Creating Fields	3
USN5	Handling Data Inconsistencies	3

Total Story Points in Sprint 1

Total =

$$2 + 1 + 3 + 3 + 3 = 12 \text{ Story Points}$$

Sprint 2

Epic 3: Data Visualization

User Story ID	Task	Story Points
USN6	Bar Chart Visualization	2
USN7	Pie Chart Visualization	2
USN8	Line Chart Visualization	2
USN9	Map Visualization	4

Epic 4: Dashboard Development

User Story ID	Task	Story Points
USN10	Developing Dashboard	5

Epic 5: Story Development

User Story ID	Task	Story Points
USN11	Developing Story	5

Total Story Points in Sprint 2

Total =

$2 + 2 + 2 + 4 + 5 + 5 = 20$ Story Points

4. Sprint Summary

Sprint Total Story Points

Sprint 1 12

Sprint 2 20

Total Story Points Completed =

$12 + 20 = 32$

5. Team Velocity Calculation

Velocity is a measure of how much work a team can complete in one sprint.

Formula

Velocity =

Total Story Points Completed / Number of Sprints

Calculation

Total Story Points = **32**

Number of Sprints = **2**

Velocity =

$32 / 2 = 16$

6. Final Result

The team's **velocity is 16 Story Points per Sprint.**

This means the team can realistically complete **approximately 16 story points of work in each sprint**, which helps in better sprint planning and workload distribution for future development cycles.